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Letter

# Gold Catalysis Enabling Furan-Fused Cyclobutenes as a Platform toward Cross Cycloadditions

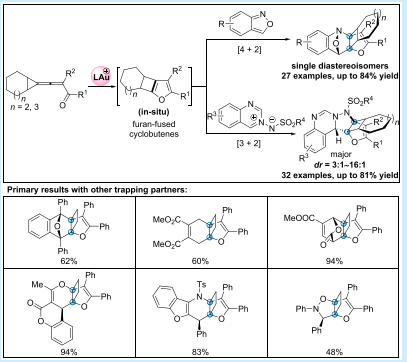
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**ABSTRACT:** The inherently strained furan-fused cyclobutenes, *in situ* generated via cycloisomerizations of allenyl ketones bearing cyclopropyl moiety under gold catalysis, have been utilized as reactive building blocks toward cross cycloadditions. The [4 + 2] and [3 + 2] annulations of these species with benzo[*c*]isoxazoles and *N*-iminoquinazolinium ylides furnish various three-dimensional cyclobutane-bridged polyheterocycles in good yields. A wide range of typically electron-deficient 1,3-dienes, heterodienes, and 1,3-dipoles can trap furan-fused cyclobutenes to afford several polycyclic architectures.




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Polyheterocycles with a cyclobutane functionality at the fused junction are important structural motif in many natural products[1] (I, II, III) and also behave as a new glycoluril diether, one kind of supramolecular precursors[2] (IV) with a wide dihedral angle on the concave face (Scheme 1A). Classical procedures, such as [2π+2π]-photocycloaddition,[3] ring-expansion of cyclopropane,[4] nucleophilic ring-closure,[5] transition-metal catalyzed cyclization,[6] and so on, have been developed for the construction of cyclobutane-containing architecturally complex molecules. However, these methods have often suffered from inconvenient reaction conditions, low selectivities, and inaccessibility of substrates. Hence, the development of a general, diversity-oriented and stereoselective strategy to access polyheterocycles with a highly strained cyclobutane at the bridged-carbons poses significant challenges and is highly desirable.

The cross cycloaddition of highly reactive organo-intermediates has emerged as a hot topic in chemistry for two decades and has been regarded as a reliable tool for rapidly constructing complex scaffolds.[7] Cyclobutenones[8] and their heterocyclic analogues[9] are among highly useful building blocks toward some cycloadditions for the construction of molecules with four-membered rings remained at the joint position (Scheme 1B). Generally, the inner C−C double in these substrates shows dramatically enhanced reactivities because of the higher strained energy of small ring systems (ca. 26.7 kcal $mol^{-1}$) relative to normal cycles. In 2010, Danishefsky[10] reported an elegant work on Diels−Alder cycloaddition employing the parent cyclobutenone as reactive dienophile to react with simple dienes and Danishefsky's dienes to afford bicycle[4.2.0]octenes with excellent selectivity (Scheme 1B, eq 1). Recently, Lu[11] disclosed an asymmetric version of Diels−Alder reaction using 3-methoxycarbonylcyclobutenone as the dienophile. Besides, Didier[12] presented the novel dipolar cycloadditions for the stereoselective formation of fused isoxazoline derivatives involving azetines and thietes, the heterocyclic analogs of cyclobutenes (Scheme 1B, eq 2). To our best knowledge, electron-rich cyclobutenes as dienophiles for the Diels−Alder reaction or as highly active species for dipolar cycloadditions are however still underdeveloped. Furthermore, homogeneous gold catalysis[13] has empowered a wide array of efficient transformations to enter structurally complex molecules based on the unique ability of gold complexes to interact with π-bonds. In this area, we[14] have developed a gold-catalyzed tandem cycloisomerization of cyclopropyl-tethered allenyl ketones to produce the key intermediates, furan-fused cyclobutenes, which could be further transformed into gold-containing 1,4-all-carbon dipoles for [4 + *n*] cycloadditions to produce various furan-based polycyclic molecules (Scheme 1C). We envisioned that the *in situ* formed furan-fused cyclobutenes with highly strained electron-donating double bond might display novel reactivities toward cycloadditions. Herein, we would like to disclose the new gold-catalyzed reactions of allenyl ketones bearing a cyclopropyl moiety, which enable the *in situ* furan-fused cyclobutenes as platform for [2 + *n*] cycloadditions for the

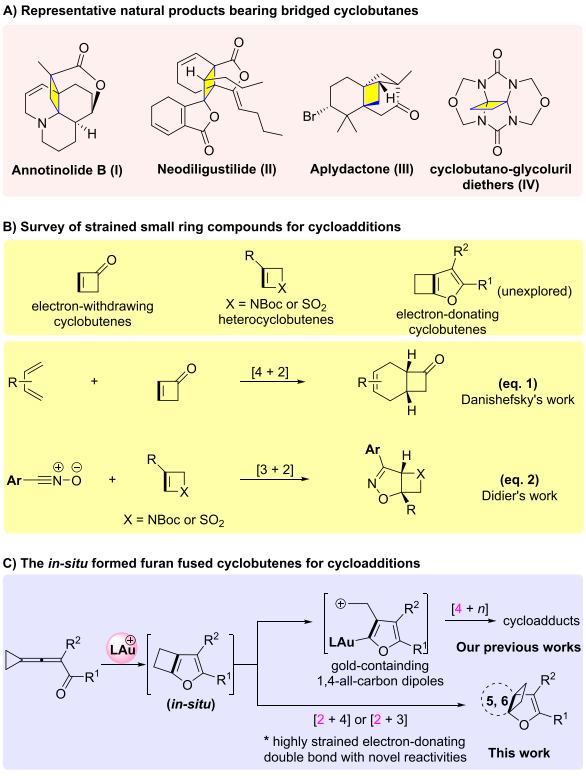
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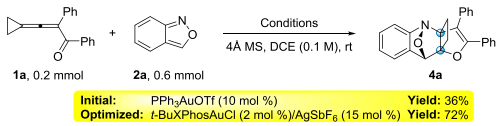
**Scheme 1. Background and Working Hypothesis**




rapid construction of valuable cyclobutane-bridged frameworks (Scheme 1C).

To fulfill our hypothesis, we initially treated the cyclopropyl-tethered allenyl ketone (**1a**) with benzo[*c*]isoxazole (**2a**) with 10 mol % of $PPh_3AuOTf$ in DCE at room temperature (rt) under $N_2$ atmosphere for 12 h. The [4 + 2] cycloadduct **4a** bearing a strained cyclobutane and *O*-bridged ring was obtained in 36% yield with excellent stereoselectivity (Scheme 2). The reaction parameters including metal catalysts,

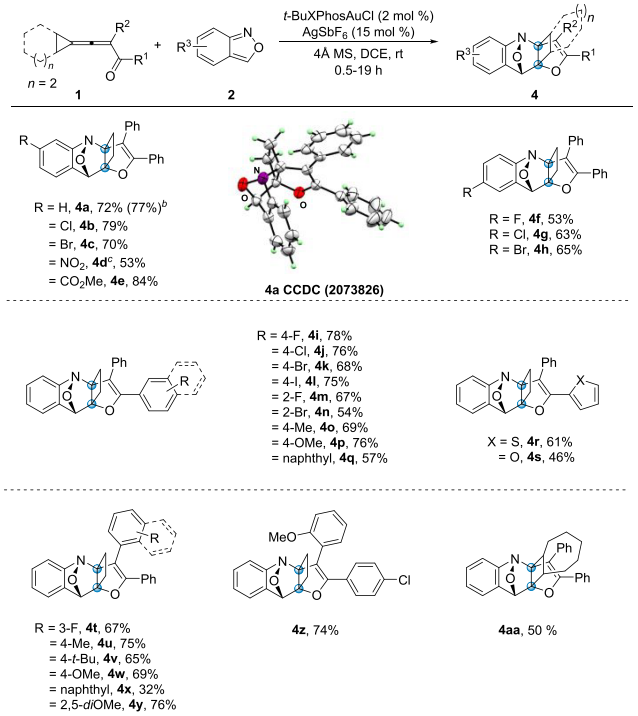
**Scheme 2. Initial Result and Optimized Conditions of [4 + 2] Cycloaddition**




additives, ligands, temperature, and solvents were systematically varied (see Table S1 in the Supporting Information (SI) for details). Other metal catalysts such as $Pd(OAc)_2$, CuI, and $Cu(OTf)_2$ were incapable of catalyzing this transformation (Table S1, entries 2–4). The examination of counteranion effects[15] indicated that the $SbF_6$ anion gave the best results, because counterions like $SbF_6^-$ have smaller affinity toward cationic gold, which may show higher reactivity in this transformation (Table S1, entries 5–8). Reducing the amount of $PPh_3AuCl$ did not result in inferior yields (Table S1, entry 9), while the use of an increasing amount of $AgSbF_6$ up to 15 mol % led to higher yields (Table S1, entry 10). Ligand effects of gold catalysts indicated that the employment of more crowded *t*-BuXPhosAuCl would greatly improve the yield of **4a** up to 72% (Table S1, entries 11–16). No better yields of product **3a** were gained by reducing or elevating the reaction temperature (Table S1, entries 18 and 19). DCE is the best choice of solvent because the reaction performed in other solvents such as DCM, chloroform, THF, toluene, acetonitrile, and DMF all gave decreased yields (Table S1, entries 20–25). Lastly, we checked the reaction using 5 mol % of $AgSbF_6$ as sole catalyst in DCM; indeed, the target **4a** could be obtained, albeit in modest yield (55%). Thus, the standard conditions were established as cyclopropyl-tethered allenyl ketone **1a** (1 equiv) and benzo[*c*]isoxazole **2a** (3 equiv) using *t*-BuXPhosAuCl (2 mol %)/$AgSbF_6$ (15 mol %) as a catalytic system in a solution of DCE (0.1 M) at rt (Scheme 2).

With the standard conditions in hand, the scope of gold-catalyzed [4 + 2] cycloaddition was investigated, and the results were summarized in Scheme 3. First, a wide range of

**Scheme 3. Substrate Scope of [4 + 2] Cycloaddition**[a]




[a]The reaction was carried out using **1** (0.2 mmol, 1.0 equiv), **2** (0.6 mmol, 3.0 equiv), *t*-BuXPhosAuCl (2 mol %), $AgSbF_6$ (15 mol %), 100 mg of 4 ÅMS in 2.0 mL of DCE at room temperature in a nitrogen atmosphere. [b]Gram-scale reaction: using **1a** (1.130 g), **2a** (1.641 g) providing **4a** (1.340 g) in 77% yield. [c]The amount of **1** (0.6 mmol, 3.0 equiv) and **2** (0.2 mmol, 1.0 equiv) was applied in this transformation.

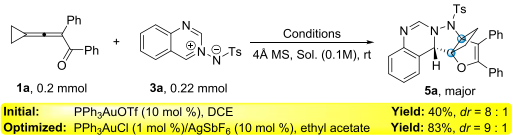
benzo[*c*]isoxazoles **2** were employed to react with cyclopropyl tethered allene ketone **1a**. Benzo[c]isoxazoles bearing Cl, Br, $NO_2$, and $CO_2Me$ groups at the C6 position of the aromatic ring reacted smoothly with **1a** to deliver the corresponding **4b–4e** in moderate to high yields. Meanwhile, the halo substituents could be suitably arranged in the C5 position of benzo[*c*]isoxazole frameworks to afford products **4f–4h** in moderate yields. Subsequently, the substrate scope of cyclopropyl tethered allene ketones **1** was then evaluated. The nature of the substituents attaching at the benzene ring of $R^1$ position had little effect on this reaction, and the desired compounds **4i–4p** were produced in 54–78% yields. 2-Naphthyl, 2-thienyl, and 2-furyl groups were also well tolerated at the $R^1$ position, furnishing the desired products (**4q–4s**) in


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moderate yields. The $R^2$ substituents with various substituted aromatic rings were all compatible for this transformation, and the cycloadducts **4t**−**4z** were obtained in modest to high yields. To our delight, a cycloheptyl fused substrate could be successfully used for the assembly of the highly bridged-skeleton **4aa** in 50% yield.

Encouraged by the aforementioned results of Diels−Alder cycloaddition involving the *in situ* formed furan-fused cyclobutenes as an ideal dienophile, we wondered whether the furan-fused cyclobutenes could be subjected to the dipolar cycloadditions. Thus, we explored the [3 + 2] cycloaddition using cyclopropyl-tethered allenyl ketone (**1a**) and quinazolin-3-ium-3-yl(tosyl)amide (**2a**) as model substrates under gold catalysis. The structurally complex cycloadduct **5a** was furnished in 40% yield with 8:1 diastereoselectivity at the initial attempt. Systematic screenings of the conditions were then carried out (see Table S2 in the Supporting Information (SI) for details). The yield of **5a** was greatly improved when the reaction was carried out using $Ph_3PAuCl$ (1 mol %)/$AgSbF_6$ (10 mol %) as the catalytic system in a solution of EA (0.1 M) at rt (Scheme 4).

**Scheme 4. Experimental Results of [3 + 2] Cycloaddition**

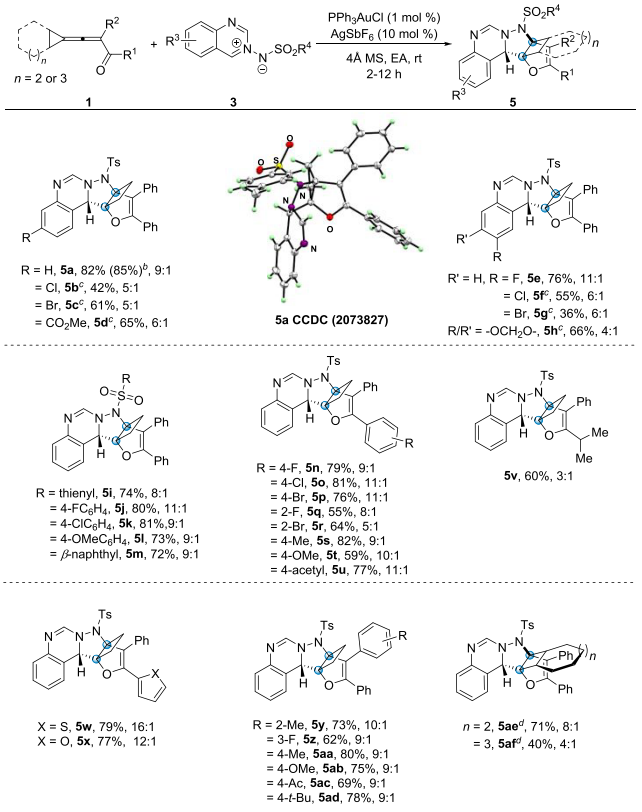



The scope of this novel [3 + 2] cycloaddition reaction was shown in Scheme 5. A variety of *N*-iminoquinazolinium ylides **3** bearing electron-withdrawing or electron-donating groups at the aromatic ring of the quinazoline backbone were well-tolerated to provide **5b**−**5h** in modest to good yields. Various substituted sulfonyl functionalities were compatible, giving **5i**−**5m** in high yields. With respect to cyclopropyl-tethered allene ketones **1**, different aromatic substituents (**5n**−**5u**), alkyl (**5v**), and heterocyclic substituents (**5w**−**5x**) were suitably arranged at the $R^1$ position, leading to high yields of products. Varying the substituents at the $R^2$ position gave the cycloadducts **5y**−**5ad** in high to good yields. The cycloheptyl or cyclooctyl-fused cyclopropyl-tethered allene ketones were also suitable substrates to afford **5ae** and **5af** in 71% and 40% yield, respectively.

To demonstrate the synthetic utilities of these cycloadditions, the transformations of the products were performed (Scheme 6). The reductive cleaving of the N−O bond of **4a** furnished quinoline derivative **6** in 89% yield. Compound **6** could be further transformed into **7a** and **7b** via selective *N*-alkylation and oxidation in high yields. Moreover, the ring-expansion reaction of **5a** in the presence of TfOH (2 equiv) afforded the seven-membered ring fused heterocycle **8** in a high yield, which could be further reduced to produce **9** under Raney Ni/$H_2$ system in an acceptable yield.

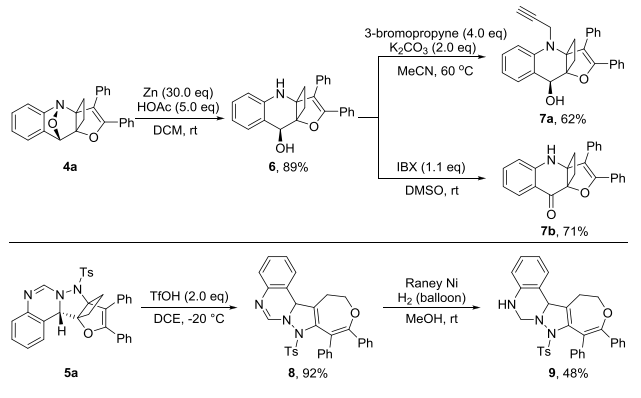
Finally, it is worth noting that the *in situ* formed furan-fused cyclobutenes could be captured by a wide range of other $\pi$-components including different 1,3-dienes, heterodienes, and 1,3-dipoles. Some representative examples are showed in Scheme 7. For instance, the highly reactive 1,3-diphenylisobenzofuran **10** was compatible to this cycloaddition to give highly bridged heterocycle **11** in 62% yield with well-defined stereoselectivity. The reactions of the electron-deficient 1,3-

**Scheme 5. Substrate Scope of [4 + 2] Cycloaddition**[a]




[a]The reaction was carried out using **1** (0.2 mmol, 1.0 equiv), **3** (0.22 mmol, 1.1 equiv), $PPh_3AuCl$ (1 mol %), AgSbF6 (10 mol %), 100 mg of 4 ÅMS in 2.0 mL of EA at room temperature under a nitrogen atmosphere. [b]Gram-scale reaction: using **1a** (0.6617 g), **3a** (0.8241 g) providing **4a** (1.161 g) in 85% yield. [c]The reaction was performed by using mixture solvent (2.0 mL 1,4-dioxane/2.0 mL DCE) at 80 °C. [d]THF was used as the solvent. [f]EA = ethyl acetate

**Scheme 6. Transformation of the Products 4a and 5a**

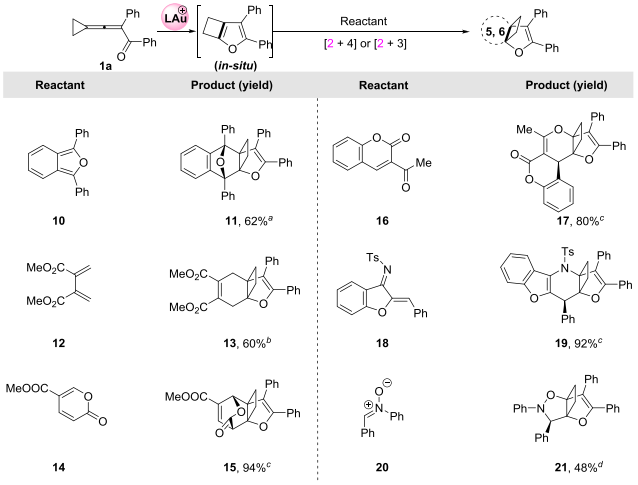



dienes **12** and **14** afforded the cycloadducts **13** and **15** stereoselectively in 60% and 94% yields, respectively. Remarkably, the furan-fused cyclobutenes could undergo the inverse-electron-demand *oxo- and aza*-Diels−Alder reaction with substrates **16** and **17** to furnish polycyclic heterocycles **17** and **19** in good yields. The nitrone **20** could react with furan-fused cyclobutene species to produce *N,O*-containing heterocycle **21** in a modest yield when THF was used as solvent. It should be noted that the reaction mode was different from the previously reported [4 + 3] cycloaddition,[14b] probably because



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**Scheme 7. Primary Results with Other Coupling Partners**




[a]The reaction was carried out using **1a** (0.1 mmol, 1.0 equiv), **10** (0.15 mmol, 1.5 equiv), *t*-BuXPhosAuCl (2 mol %), AgTFA (15 mol %), 100 mg of 4 ÅMS in 2.0 mL of DCE at room temperature in a nitrogen atmosphere. [b]Using **1a** (0.2 mmol, 1.0 equiv), **12** (0.6 mmol, 3 equiv), $PPh_3AuCl$ (2 mol %), AgTFA (15 mol %), 100 mg of 4 ÅMS in 2.0 mL of DCE. [c]Using **1a** (0.2 mmol, 1.0 equiv), **14**, **16** or **18** (0.2 mmol, 1.0 equiv), $PPh_3AuOTf$ (10 mol %), 100 mg of 4 ÅMS in 2.0 mL of DCE. [d]Using **1a** (0.2 mmol, 1.0 equiv), **20** (0.6 mmol, 3.0 equiv), $AgSbF_6$ (10 mol %), 100 mg of 4 ÅMS in 2.0 mL of THF.

the $\pi$-Lewis acidity of metal catalysts was weakened by O-containing solvent and the ring-opening reaction of *in situ* formed furan-fused cyclobutene intermediate to produce the 1,4-dipole might be inhibited.

In conclusion, we reported the generation of furan-fused cyclobutenes under mild gold catalysis, which could be used as a platform for [2 + *n*] cycloadditions. A wide variety of polycyclic heterocycles bearing cyclobutane at the bridged-carbons were efficiently constructed in modest to good yields with good stereoselectivity. The selection of other $\pi$-component traps to create structurally complex molecules is currently under investigation in our laboratory.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c01079.

Experimental procedures and copies of $^1H$ and $^{13}C$ NMR spectra for all new compounds (PDF)

### Accession Codes

CCDC 1962639 and 2073826−2073828 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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### Notes

The authors declare no competing financial interest.

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